

IS470 Research proposal sketch

Explicit Recommendation incorporating uncertainty and abstention

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1 Introduction and research area

Informally, our research area can be summed up as: recommending things is easy, recommending things we think people will like is hard, recommending things people will like *and knowing when we cannot* is harder still. Recommendation research generally falls into two broad categories, *explicit feedback* and *implicit feedback*. Implicit feedback deals with whether a user will interact with some item in the future, while explicit feedback deals with whether a user will like a particular item, conditionally given that they will interact. Take the example of movie recommendation. An implicit-feedback system tries to answer the question "will this user watch this movie?" while an explicit-feedback system asks "if the user watches this movie, what rating will they give it?".

We extend the above question for explicit-feedback systems and ask "if the user watches this movie, how confident that this is the rating will they give it? and if we are not confident, can we not recommend them anything?"

This research project aims to improve explicit recommendation systems through joint training and incorporation of recommendation uncertainty with abstention. We enable the system to achieve better performance by having a mathematical structure for both the prediction of how much a user will like an item and for how unsure it is of its own prediction.

2 Research goal

We want to test if an uncertainty aware explicit-feedback recommender can be trained, and if abstaining on unreliable predictions yields a better accuracy-coverage trade-off than abstention heuristics that do not use a learned uncertainty.

2.1 Hypothesis

A jointly-trained uncertainty model should outperform naive abstention baselines across all possible abstention percentages. At zero abstention, the uncertainty-trained model should still beat naive abstention methods for plain matrix factorisation.

A Priori, understanding what the system's prediction is and also how uncertain it is of its own recommendation is useful to understand explicit feedback recommendation. We want to find some mathematical way to understand this uncertainty's structure and *hypothesise that this is also a low-rank matrix*. The rationale, loosely, is as follows: one of the rating matrix's low-rank interpretations is that across many people we are not as idiosyncratic as we think; therefore, everyone's wants can be explained by a few latent factors. We can then be relatively confident when we recommend something for someone who exhibits behavior that is aligned to the factors, but are unable to confidently recommend to someone else who exhibits behavior outside of the latent factors. A fuzzy example is if Timmy is a hardcore star wars fan, we can be rather

confident of their likes and dislikes based on other star wars fans, but we are less confident of Bob, who might exhibit behaviors that cannot be inferred from other movie watcher's habits.

2.2 Motivation

Our main motivation is that, with uncertainty, the RMSE will improve. We are basing this on previous results. We know that by performing joint training of explicit feedback recommender systems, we are able to achieve near state of the art performance [1]. It is then not unfounded to posit that for uncertain items, letting the model abstain from recommending items will improve the selective RMSE score. We can also model this as the model having one more signal. Having an uncertainty signal during training will expand the ability for our model to match the underlying real-world behaviour. This is saying that training a model with an uncertainty bound should enhance the RMSE even when we force it to not abstain on any question, since uncertainty still serves a valuable perspective.

3 Formal problem statement

Let $u_1, \dots, u_n$ be users, indexed by $i \in [n]$, and $v_1, \dots, v_m$ items, indexed by $j \in [m]$. Ratings take values in $\mathcal{R} = \{1, \dots, 5\}$. The complete rating matrix $R \in \mathcal{R}^{n \times m}$ is observed only on $\Omega \subseteq [n] \times [m]$ with $|\Omega| \ll nm$; the remaining entries are missing. We split Ω into disjoint $\Omega_{\text{train}}, \Omega_{\text{val}}, \Omega_{\text{test}}$ in proportion 70/20/10.

A conventional explicit-feedback recommender is a function $f(i, j) = \hat{S}_{ij}$ returning a predicted rating. We instead train a model to predict both the rating and its uncertainty.

$$f(i, j) = (\hat{S}_{ij}, \hat{U}_{ij}), \quad (1)$$

where larger $\hat{U}_{ij}$ denotes lower confidence. We require of $\hat{U}$ only that it *order* entries by reliability, not that it be calibrated.

For an abstention rate $p \in [0, 1)$, the model declines to answer on the fraction p of test entries carrying the largest $\hat{U}$, retaining

$$\Omega_p(f) \subseteq \Omega_{\text{test}}, \quad |\Omega_p(f)| = \lceil (1 - p) |\Omega_{\text{test}}| \rceil, \quad (2)$$

and is scored on what remains:

$$\text{RMSE}(f, p) = \sqrt{\frac{1}{|\Omega_p(f)|} \sum_{(i,j) \in \Omega_p(f)} (\hat{S}_{ij} - R_{ij})^2}. \quad (3)$$

At $p = 0$ this reduces to ordinary test RMSE.

4 Current experimental design

4.1 Current model

We will evaluate a naive matrix decomposition recommender system,

$$S = AB^\top, \quad U = CD^\top, \quad (4)$$

with $A \in \mathbb{R}^{n \times k}$, $B \in \mathbb{R}^{m \times k}$ and likewise for C, D . Both are fitted on the observed entries, the only entries at which a residual can be formed:

We co-train both together using the following loss:

$$\mathcal{L}(A, B, C, D) = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} (S_{ij} - R_{ij})^2 \exp(-U_{ij}) + \lambda \sum_{(i,j) \in \Omega} U_{ij} \quad (5)$$

4.2 Baselines and datasets

A low-rank $\hat{U}$ is only interesting if it abstains better than signals requiring no training. We compare against IGMF [2], SoftImpute [3], AutoRec [4], plain matrix factorisation [5], and UAIMC [1], which is the nearest prior work and already produces an uncertainty estimate.

Since none of these methods has an abstention mechanism, we will implement two naive abstention mechanisms as a baseline. The first is random abstention; the model just randomly decides what not to answer. The second is abstention on lower support ratings, where, if the model sees that we have very little ratings for the user, the model then chooses not to answer.

We additionally compare full test RMSE at $p = 0$ against plain matrix factorisation, matched on k , on μ_S and on search budget, to test if the uncertainty term improves the rating head even when abstention is disabled. Hyperparameters for this comparison alone are selected on validation full RMSE, so as not to advantage the uncertainty-equipped model by construction.

We evaluate on the following datasets:

- MovieLens-100K
- MovieLens-1M
- MovieLens-25M
- Douban
- Amazon Games

4.3 Experiments

	ML-100K				ML-1M				ML-25M				Douban				Am. Games			
Method	0	.05	.10	.20	0	.05	.10	.20	0	.05	.10	.20	0	.05	.10	.20	0	.05	.10	.20
Ours ($\hat{U}$)																				
UA-IMC ($\hat{U}$)																				
IGMC (rand)																				
IGMC (supp)																				
SoftImpute (rand)																				
SoftImpute (supp)																				
AutoRec (rand)																				
AutoRec (supp)																				
Plain MF (rand)																				
Plain MF (supp)																				

Table 1: Selective RMSE by abstention rate p . Rows sharing a method are the same trained model scored under two abstention rules, so the $p = 0$ columns agree within each pair; the grid covers 30 training runs.

5 Broader problem space and extensions

The above described experiment is brazenly simple and there are many areas where things could not work for our model and the hypothesis. The biggest one is that, for now, we have abstention per recommendation, whereas an abstention on the user is the more widely deployed idea[cold start soft impute]. We also might change the loss function.

We also need to study if the hypothesis U is low rank holds. Supposing it is not, we can open up our investigation to better understand if there are other structures that bound it.

References

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